

SARDAR PATEL UNIVERSITY

B.Sc.(CA & IT) (INTEGRATED) (SEM - IV) EXAMINATION MARCH/APRIL – 2019

PS04FIIT01 : COMPUTER BASED NUMERICAL & STATISTICAL METHOD

Date: 18/03/2019, Monday

Time: 02:00 to 05:00 PM

Total Marks : 70

Q.1 Multiple choice questions:

02-00 To 05-00 PM

[10]

- 1 For real root of an equation $x^3 - 2x - 5 = 0$, the root lies between
(a) 0 and 1 (b) 2 and 3 (c) 1 and 2 (d) none of them
- 2 The number 0.01850×10^3 has _____ significant digits
(a) 3 (b) 4 (c) 5 (d) 6
- 3 y depends on x can be written as _____
(a) $f(x)$ or y_x (b) $f(xy)$ (c) $f(yx)$ (d) none of above
- 4 _____ is not a type of interpolation method.
(a) Forward difference (b) backward difference
(c) Newton's divided difference (d) moving average method
- 5 The second order differences is denoted by _____.
(a) Δ^2 (b) ∇^3 (c) ϕ^2 (d) y^3
- 6 The system of linear equation $AX = B$ can solved by matrix inversion method only if...
(a) $A \neq 0$ (b) $|A| \neq 0$ (c) $|A| = 0$ (d) A is symmetric
- 7 Rate of change of distance with respect to time represents....
(a) Acceleration (b) Speed (c) Pressure (d) None of these
- 8 A forecast with a time horizon of about 3 months to 3 years is typically called a _____.
(a) long-range forecast (b) medium-range forecast
(c) short-range forecast (d) weather forecast
- 9 Irregular variations in a time series are caused by:
(a) lockouts and strikes (b) epidemics
(c) floods (d) all the above
- 10 Secular trend is indicative of long-term variation towards:
(a) increase only (b) decrease only
(c) either increase or decrease (d) none of the above

Q.2 Attempt any ten out of twelve.

[20]

1. Define Relative error and absolute error.
2. Describe the stopping rules to obtain approximate solution for given non-linear equations.
3. Find the next iterative value of the root of using secant method, if the initial guesses are 3 and 4.
4. Define Interpolation & Extrapolation
5. Write different methods of Interpolation.
6. Explain central difference method.
7. If x lies in the upper half of the table and if $x_k < x < x_{k+1}$, then what is $\frac{dy(x)}{dx}$ and $\frac{d^2y(x)}{dx^2}$?
8. List only various direct and iterative methods.

9. List the utilities of time series.
10. Differentiate between linear and non linear trend.
11. List the component of Time series.
12. Write a note on Survey Method of Forecasting.

Q.3 (A) $x^3 - x - 4 = 0$. Solve this polynomial using secant method. [5]

Q.3 (B) Write algorithm for bisection method. [5]

OR

Q.3 (A) Write algorithm for Regula Falsi method. [5]

Q.3 (B) $2x - \cos x - 3 = 0$, Take $a=1.5$ solve this using iterative method. [5]

Q.4 (A) Using an appropriate formula for interpolation estimate the no. of students who obtained < 45 marks from the following. [5]

Marks	0-40	40-50	50-60	60-70	70-80
No. of students	31	42	51	35	31

Q.4 (B) Given the table of values as (use Lagrangian method) [5]

x	0	1	2	3
y(x)	0	2	8	27

Find $y(2.35)$.

OR

Q.4 (A) Apply Newton's divided difference method to find the no. of persons getting Rs. 6 from the following data. [5]

Income per day	3	5	7	8	10
No. of persons	180	154	120	110	90

Q.4 (B) The value of x and $y = f(x)$ are given below [5]

x	5	6	9	11
f(x)	12	13	14	16

Find value of x when $f(x) = 15$.

Q.5 (A) Solve the following system of equations using matrix inversion method. [5]

$$2x_1 + x_2 + x_3 = 10$$

$$3x_1 + 2x_2 + 3x_3 = 18$$

$$x_1 + 4x_2 + 9x_3 = 16$$

(B) The distance (s) covered as a function of time (t) by an athlete during his/her run for the 50 meter race is given in the following table [5]

Time(Secs.)	0	1	2	3	4	5	6
Distance(Mts.)	0	2.5	8.5	15.5	24.5	36.5	50

Determine the speed of the athlete at $t = 5$ and $t = 4.5$ seconds.

OR

- (A) Solve the following system of equations using Gauss-Seidel method. [5]

$$5x_1 + 2x_2 + x_3 = 12$$

$$x_1 + 4x_2 + 2x_3 = 15$$

$$x_1 + 2x_2 + 5x_3 = 20$$

- (B) Given the following table [5]

x	3	4	5	6	7	8
y	4	6.6	7.7	9.0	10.5	12.2

Use backward difference formula to compute $\frac{dy}{dx}$ at $x = 7$ and $x = 7.5$.

- Q.6(A) Explain Extrapolation model in forecasting. [5]

- Q.6(B) Obtain the trend from the time series given below by method of moving average of [i] 3 years [5]
[ii] 5 years

Year	1961	1962	1963	1964	1965	1966	1967	1968	1969	1970
Y	500	540	550	530	520	560	600	640	620	640

OR

- Q.6(A) Obtain seasonal indices using simple average method. [5]

Year	Q-I	Q-II	Q-III	Q-IV
1974	30	81	62	119
1975	33	104	86	171
1976	42	153	99	221
1977	56	172	129	235

- Q.6(B) Obtain seasonal indices using ratio to moving average. [5]

Year	Q-I	Q-II	Q-III	Q-IV
1970	25	30	21	32
1971	27	28	25	34
1972	22	27	21	30
1973	24	25	20	33

— X —
(3)

